

YASWANTH REDDY BALAM

AI Engineer

☎ +91-8179935501 ✉ yrb1211@gmail.com 🔗 [linkedin.com/in/yaswanth-reddy-balam](https://www.linkedin.com/in/yaswanth-reddy-balam) 🐙 github.com/yaswanthreddy3

Summary

AI Engineer with production experience building Python backend services for AI systems and hands-on work with LLM-based agentic workflows. Built an MCP-based LLM agent with tool orchestration, deduplication, and context management; shipped a production computer vision pipeline (YOLO11, 80% mAP) behind an async FastAPI/Docker service. Comfortable across the full stack: model integration, backend APIs, data pipelines, and containerized deployment.

Technical Skills

Languages: Python, C++, SQL, Git

GenAI & Agentic AI: LLMs, Model Context Protocol (MCP), Tool Calling, Prompt Engineering, LangChain, LangGraph, AI Agents, Context Management

Backend & APIs: FastAPI, REST APIs, JSON, OpenAPI/Swagger, ONNX Runtime, Async Python

Computer Vision: YOLO11, OpenCV, Object Detection, Data Augmentation, Model Optimization

Data & MLOps: MLflow, PostgreSQL, MinIO, Redis, Docker, Docker Compose, Kong API Gateway, Experiment Tracking, Linux

Experience

AI/ML Engineer

September 2025 – Present

Spurring Ventures

Bengaluru

- Built an async FastAPI inference service integrating a fine-tuned YOLO11 detection model, handling video ingestion from MinIO, adaptive frame sampling, and structured result logging to PostgreSQL.
- Designed REST APIs for image/video inference and optimized payload delivery (Base64 to signed URLs), cutting response time by ~42%; containerized the full stack with Docker Compose, Kong API Gateway, and MLflow for experiment tracking.
- Collected and annotated a ~17K-image dataset and fine-tuned YOLO11 (s/m/l) from COCO-pretrained weights, tuning augmentation and hyperparameters to reach 80% mAP on person detection.

Intern

March 2025 – August 2025

Harman International (Intel Client)

Bengaluru

- Automated Intel S3/S4 sleep-wake stress testing with Python, eliminated manual execution, and accelerated root-cause diagnosis through log analysis.
- Collaborated with cross-functional hardware and software teams to streamline validation workflows and improve test reliability.

Projects

FoodBot CLI – Agentic LLM Assistant | *Model Context Protocol (MCP), Ollama, LLM Agents, Python*

- Built a terminal-based LLM agent using MCP for tool orchestration, implementing tool-call deduplication, context compression to manage token limits, and human-in-the-loop confirmation before executing actions – covering core agentic AI patterns of planning, tool execution, and controlled automation.

Construction Site Person & Vehicle Detection | *PyTorch, YOLO11, ONNX, FastAPI, Docker, PostgreSQL, MinIO*

- Engineered an end-to-end CV pipeline – dataset annotation, YOLO11 fine-tuning, ONNX export, async FastAPI inference, and Docker deployment – adding hard-negative samples to cut background false positives.

Certifications

- **Machine Learning Specialization** – DeepLearning.AI & Stanford Online
- **Microsoft Azure AI Fundamentals**

Education

B.E. in Information Science

SJC Institute of Technology

2020 – 2024

Chikballapur